

696. Count Binary Substrings

Solved 

Easy

 Topics

 Companies

 Hint

Given a binary string `s`, return the number of non-empty substrings that have the same number of `0`'s and `1`'s, and all the `0`'s and all the `1`'s in these substrings are grouped consecutively.

Substrings that occur multiple times are counted the number of times they occur.

Example 1:

Input: `s = "00110011"`

Output: 6

Explanation: There are 6 substrings that have equal number of consecutive 1's and 0's: "0011", "01", "1100", "10", "0011", and "01".

Notice that some of these substrings repeat and are counted the number of times they occur.

Also, "00110011" is not a valid substring because all the 0's (and 1's) are not grouped together.

Example 2:

Input: `s = "10101"`

Output: 4

Explanation: There are 4 substrings: "10", "01", "10", "01" that have equal number of consecutive 1's and 0's.

Constraints:

- `1 <= s.length <= 105`
- `s[i]` is either `'0'` or `'1'`.

Python:

class Solution:

def countBinarySubstrings(self, s: str) -> int:

prev,curr,ans=0,1,0

for i in range(1,len(s)):

if s[i]==s[i-1]:

curr+=1

else:

ans+=min(prev,curr)

prev,curr=curr,1

return ans+min(prev,curr)

JavaScript:

```
var countBinarySubstrings = function(s) {  
    let prev = 0, curr = 1, ans = 0;  
  
    for(let i = 1; i < s.length; i++){  
        if(s[i] === s[i-1]) curr++;  
        else{  
            ans += Math.min(prev, curr);  
            prev = curr;  
            curr = 1;  
        }  
    }  
    return ans + Math.min(prev, curr);  
};
```

Java:

```
class Solution {  
    public int countBinarySubstrings(String s) {  
        int prev=0,curr=1,ans=0;  
        for(int i=1;i<s.length();i++){  
            if(s.charAt(i)==s.charAt(i-1)) curr++;  
            else{  
                ans+=Math.min(prev,curr);  
                prev=curr;  
                curr=1;  
            }  
        }  
        return ans+Math.min(prev,curr);  
    }  
}
```